

ML Specification Report — playground-series-s4e11

Depression Prediction · our from-scratch agent (GBDT pool + tree-search v3)

Figures grounded in the competition's `facts.json`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

Overview

The task is to classify 93,800 survey respondents as depressed or not from demographic/lifestyle features (metric: Accuracy). We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) refined by a four-tier ablation and a 60-node tree search that ends on an 8-member blend. Because Accuracy needs a hard 0/1 label, every tier also optimizes a probability **decision threshold**. Our champion scores **CV Accuracy 0.940235**; the NVIDIA reproduce-agent, distilling the essence of a public XGB/CatBoost/LGB voting kernel, reaches 0.9399 — a dead heat, with our agent nudging ~0.0003 ahead.

Why it matters. Early identification of depression risk from survey and lifestyle factors can route scarce mental-health resources to those most likely to benefit — as a screening aid, not a diagnosis.

Data

Purpose of Data. Predict whether a survey respondent is depressed — a binary-classification Playground Series episode. **Data Format** is clean **tabular CSV**, a mix of numeric and categorical columns. **Data Volume** is **140,700 training rows / 93,800 test rows**, with 18 raw predictive features once `id` and the target are removed.

Data Quality is high on the structural axis — **0 duplicate rows** and no train↔test mean shift (all eight numeric features within $\pm 0.52\%$) — but two real wrinkles shape modelling. First, **class imbalance**: only **18.2% of respondents are depressed** (`Depression=1`; 25,567 / 140,700), which drives the decision-threshold tuning below. Second, the missing values are **structured, not random**: the *Working Professional or Student* split creates disjoint feature blocks. Academic Pressure / CGPA / Study Satisfaction are ~80% null (112,802–112,803 rows) because they only apply to students, while Work Pressure / Job Satisfaction are ~20% null (27,910–27,918 rows) because they only apply to professionals. The strongest EDA signals, all consistent with domain intuition, were:

- **Age** is the dominant predictor (Pearson $r \approx -0.56$); younger respondents (students) are far more depressed.
- The **Working Professional vs Student** split is the #1 structural feature: **students (19.8% of rows) have a 58.6% depression rate vs professionals (80.2%) at 8.2%**.
- **Suicidal thoughts** is highly predictive: *Yes* → 31.8% depression vs *No* → 4.9%.

- **Academic Pressure** correlates strongly with depression among students ($r \approx +0.48$); **Financial Stress** correlates in both groups ($r \approx +0.23$).
- High-cardinality text columns are noisy: **City** (98 values) and **Degree** (115 values) carry unseen categories in test, and *Sleep Duration* / *Dietary Habits* have rare junk categories (e.g. "50-75 hours", "Mealy").

Annotation Guidelines. The label is `Depression` $\in \{0, 1\}$ (1 = depressed). Submissions are **hard 0/1 labels** scored by Accuracy — a thresholded decision, not a raw ranking.

Feature Set. From the 18 raw columns we engineer **28 features**, grouped below:

Group	Features
Binary encodings	<code>is_student</code> , <code>suicidal_thoughts</code> , <code>family_history</code> , <code>is_male</code>
Age	<code>age_decade</code> , <code>is_young</code>
Cleaned categoricals	Sleep Duration $\rightarrow$ 4 groups, Dietary Habits $\rightarrow$ 3 groups + other
Interactions	<code>pressure</code> (academic+work), <code>satisfaction</code> , <code>stress_x_pressure</code> , <code>suicidal_x_student</code> , <code>age_x_pressure</code>
Label-encoded high-cardinality	<code>City</code> , <code>Profession</code> , <code>Degree</code>
Structured-NaN handling	absent professional/student blocks filled with 0

Splitting strategy. A single canonical **Stratified 5-Fold CV (seed = 42)**, shared across all four tiers so scores are directly comparable. A deliberate leakage audit — carried over from the s4e1 surname target-encoding lesson — confirmed the safeguard is unnecessary here: the feature set uses **only label encoding** for the high-cardinality categoricals (City / Profession / Degree) and **no target encoding of any kind**, so there is no analogous fold-safety bug to correct and this run is honestly comparable to the February baseline. The **leaderboard for this run is not recorded** (`facts.json` `leaderboard: null`); scores here are **CV-only**, with the February prior-season submission (CV 0.93738 $\rightarrow$ Public LB 0.94093 $\rightarrow$ Private LB 0.93939) anchoring CV trustworthiness.

Models & Architecture

Purpose of Architecture. A binary depression classifier that maximizes thresholded Accuracy. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM as the primary learner, with XGBoost and CatBoost members, combined by a weighted blend; the champion is an **8-member blend** discovered at tree-search node #26.

Each member takes the same **Input Format** — a numeric feature matrix with categoricals encoded (binary / label / cleaned-group) — of **Input Dimension 28 features** per row (a 1-D vector; there is no spatial/sequence structure).

Architecture Description. Each GBDT member is an additive ensemble of decision trees grown by histogram-based, leaf-wise gradient boosting. Two representative LightGBM configurations appear: the February baseline (`num_leaves` = 63, ~1266 boosting iterations) and the Optuna-tuned member (`num_leaves` = 48, `max_depth` = 3, `learning_rate` $\approx$ 0.070, `min_child_samples` = 51). The blend layer

sits on top as a convex weighted average of the members' out-of-fold predictions, with weights found by Dirichlet sampling (k=800, seed=42) plus coordinate-ascent refinement, maximizing OOF threshold-optimized Accuracy. **Model Complexity** is therefore best expressed as trees × leaves rather than a dense parameter count; the champion blends 8 such members, dominated by node #10 (an LGB variant) at weight 0.603.

Training procedures

Training proceeds as a **four-tier ablation** on the shared 5-fold split, each tier adding capability and measured on the same OOF scale:

Tier	Configuration	OOF Accuracy
1	generic 3-model blend (18 raw label-encoded columns)	0.93955
2	skill pipeline: 28 features, 3-model LGB+XGB+CAT blend	0.939488
3	+ linear iteration (is_unbalance ablation / Optuna 50-trial tune / seed-bag), 5-way blend	0.9399
4	+ tree-search harness_v3 (60 nodes, node #26 = 8-member blend)	0.940235

(Note the honest non-monotonicity: the generic tier-1 blend on raw columns edged the tier-2 engineered blend by ~0.0001; the engineered features paid off only once the tier-3/tier-4 tuning and search were layered on.)

The **Loss Function** is binary log-loss (LightGBM `binary` objective) with model selection on threshold-optimized Accuracy; the `is_unbalance` class-weighting was ablated and *removing* it was better ($\Delta -0.001230$ when enabled) — a notable result, since this was the first test of the AUC-era `is_unbalance` prior on a *threshold* metric, and the prior direction transferred. The **Optimization Algorithm** is gradient boosting (histogram, leaf-wise) — **not** SGD/ADAM — with blend weights optimized by Dirichlet sampling / coordinate-ascent. The **Learning Rate** is the boosting shrinkage (LGB `learning_rate` = 0.05 baseline, ≈ 0.070 for the Optuna-tuned member); there is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

Training Duration for the tier-4 search was ~1,607 s of wall-clock over 60 evaluated nodes (the search log records 1,656.1 s total, 0 failed / 0 dedup rejections, `stop_reason` = "hard budget cap reached (60/60 evaluated nodes)"); the Optuna tune was 50 TPE trials / 437.7 s, and the February baseline LGB fit in ~1266 iterations / 171.5 s. **Training Memory** was not explicitly capped — a 140.7k × 28 float matrix is trivial on the single arm64 CPU machine, peak not separately measured. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-season experience injection** — the `is_unbalance` / Optuna fold-proxy tune / seed-bag priors were carried in from earlier seasons' `experience.md`, with the `is_unbalance` prior tested for the first time on a threshold metric here. **Data Augmentation** is N/A (*tabular*); the analogues are feature engineering (the 28 features) and seed-bagging (Optuna-tuned LGB re-fit at seed 2024).

Reproducibility Standards are strict: fixed seeds (fold seed = 42; Optuna and seed-bag seeds logged) and an OOF digit-reproduction gate that passed before the submission was built — all 8 champion members were retrained from their tree configs to $\max|\Delta\text{OOF}| \leq 5.53\text{e-}14$ vs the search's cached OOFs (well under

the 1e-9 tolerance), the weight-search replay recovered blend Accuracy 0.940235 exactly, and the decision threshold was re-derived at 0.49. This was a **native v3 run (no replay needed)**.

Inference procedures

Decision Threshold is **0.49** — the OOF-optimized probability cut the champion node #26 blend re-derived at submission time. Unlike the s4e1 AUC task (which submits raw probabilities), Accuracy requires a hard 0/1 label, so the 93,800 test probabilities were thresholded at 0.49 before writing the submission. (For reference, the February baseline had tuned a higher 0.75 cut on a single is_unbalance LGB; the blended, no-imbalance champion settled near 0.49.) **Inference Duration** was not separately profiled; GBDT scoring of the 93,800 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Classify each test respondent as depressed or not. **Performance Metrics.** Accuracy is the sole competition metric; our tier-4 champion reaches **CV OOF Accuracy 0.940235**, climbing 0.93955 (tier 1) → 0.939488 (tier 2) → 0.9399 (tier 3) → 0.940235 (tier 4). Reference baselines: a majority-class predictor scores 0.81829, and the February prior-season run posted CV 0.93738 / Public LB 0.94093 / Private LB 0.93939 — reference only, as this run is CV-only and its **own leaderboard result is not recorded**.

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which distills the essence of the strongest public kernel:

Agent	Approach	CV Accuracy	Note
Our agent	from-scratch GBDT pool + tree-search	0.940235	nudges ahead
NVIDIA	reproduces (essence) <code>mayukh18/feature-eng-cv-voting-xgb-catb-lgb</code>	0.9399	tie

The result is effectively a **tie** (our 0.9402 vs NVIDIA 0.9399, +~0.0003 to our agent). Unlike the earlier S3/S4-era churn problem (s4e1) where the well-optimized public kernel kept a small edge, here our from-scratch pipeline matches and marginally exceeds the reproduce-agent — consistent with the newer-season pattern where the reproduce-vs-originate gap collapses to a dead heat.